Project Executable Files

Date	15 March 2026
Team ID	TM-OPTICROP-04
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	3 Marks

Step 1: Submission Checklist

S.No	Item to Submit	Submitted (Yes/No/NA)
1	Complete source code	Yes
2	README / Setup Guide	Yes
3	requirements.txt	Yes
4	Database schema / CSV dataset	Yes
5	Environment config file (.env)	Yes
6	Deployed application URL	Yes
7	Executable binary	NA
8	Dockerfile	Yes
9	Test files and test results	Yes
10	Demo video walkthrough	Yes

Step 2: File / Folder Structure

```
/opticrop-root
/data - Crop_recommendation.csv
/models - crop_model.joblib, scaler.joblib
/templates - index.html, result.html
/static - style.css, script.js
/src - app.py, train.py
README.md - Setup and run instructions
requirements.txt - Python dependencies
Dockerfile - Containerization config
```

Step 3: Deployment / Access Details

Field	Details
Hosted / Deployed URL	https://opticrop-demo.example.com
Login Credentials	demo_user / demo_password123
Platform / Hosting	IBM Cloud / Render
Repository Link	https://github.com/team-opticrop/opticrop
Demo Video Link	https://youtu.be/opticrop-demo-link

Step 4: Run Instructions

```
Pre-requisites: Python 3.8+, Anaconda, Git
Step 1: git clone https://github.com/team-opticrop/opticrop
```

```
Step 2: pip install -r requirements.txt
Step 3: copy .env.example to .env
Step 4: Run training (optional): python train.py
Step 5: Start the application: python app.py
Step 6: Open browser at http://localhost:5000
```

Step 5: Known Issues / Limitations

S.No	Known Issue / Limitation	Workaround / Status
1	Initial model loading latency	Cache implemented on Flask server
2	Limited to 22 supported crops	Future expansion planned for regional varieties
3	Requires internet for CSS CDNs	Offline fallback provided in static folder